

STAT 440 Milestone 2: EDA & Initial Models

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1 Overview

The purpose of this project is to explore the relationship that mental health and other life style factors that can affect a college student's GPA. This dataset collects data on GPA, stress level, student habits, and other life style factors. The main key variables in this data set is the response variable *GPA* and the explanatory variable *study hours per day*.

Note: The dataset in this initial data analysis was changed due to the previous data being a synthetically generated Kaggle dataset, thus providing no real relationships in the data. This new data has many of the same variable as before and can be used to answer our original research question. The new dataset can be found [here](#).

1.1 Package Setup

```
if (!requireNamespace("pacman", quietly = TRUE)) install.packages("pacman")
pacman::p_load(tidyverse, janitor, skimr, GGally, broom, patchwork, knitr, kableExtra)
set.seed(440540)
```

1.2 Data Import and Cleaning

```
raw <- readr::read_csv("../data/raw/student_lifestyle_dataset.csv") |> janitor::clean_names()

dat <- raw |>
  dplyr::select(-student_id)

dat <- dat |>
  dplyr::mutate(
    stress_level = dplyr::case_when(
      stress_level == "Low" ~ 1,
      stress_level == "Moderate" ~ 2,
      stress_level == "High" ~ 3
    )
  )
```

1.3 Descriptive Statistics

```
summary_data <- dat |>
  dplyr::select(where(is.numeric)) |>
  skimr::skim()

# First half of kable
summary_data |>
  dplyr::select(
    skim_type,
    skim_variable,
    n_missing,
    complete_rate,
    numeric.mean,
    numeric.sd,
  ) |>
  knitr::kable()

# Second half of kable
summary_data |>
  dplyr::select(
    skim_variable,
    numeric.p0,
    numeric.p25,
    numeric.p50,
    numeric.p75,
    numeric.p100
  ) |>
  knitr::kable()
```

Table 1: Summary statistics for numeric variables.

skim_type	skim_variable	n_missing	complete_rate	numeric.mean	numeric.sd
numeric	study_hours_per_day	0	1	7.47580	1.4238884
numeric	extracurricular_hours_per_day	0	1	1.99010	1.1558547
numeric	sleep_hours_per_day	0	1	7.50125	1.4609485
numeric	social_hours_per_day	0	1	2.70455	1.6885141
numeric	physical_activity_hours_per_day	0	1	4.32830	2.5141101
numeric	gpa	0	1	3.11596	0.2986735
numeric	stress_level	0	1	2.36600	0.7275360

Table 2: Summary statistics for numeric variables.

skim_variable	numeric.p0	numeric.p25	numeric.p50	numeric.p75	numeric.p100
study_hours_per_day	5.00	6.3	7.40	8.70	10
extracurricular_hours_per_day	0.00	1.0	2.00	3.00	4
sleep_hours_per_day	5.00	6.2	7.50	8.80	10
social_hours_per_day	0.00	1.2	2.60	4.10	6
physical_activity_hours_per_day	0.00	2.4	4.10	6.10	13
gpa	2.24	2.9	3.11	3.33	4
stress_level	1.00	2.0	3.00	3.00	3

1.4 Exploratory Visualizations

```
# GGally::ggpairs(dplyr::select(dat, where(is.numeric)))

GGally::ggpairs(
  dplyr::select(dat, where(is.numeric)),
  lower = list(
    continuous = GGally::wrap("points", alpha = 0.03)
  )
) +
  ggplot2::theme(
    axis.text = ggplot2::element_text(size = 5),
    strip.text = ggplot2::element_text(size = 4)
  )
```

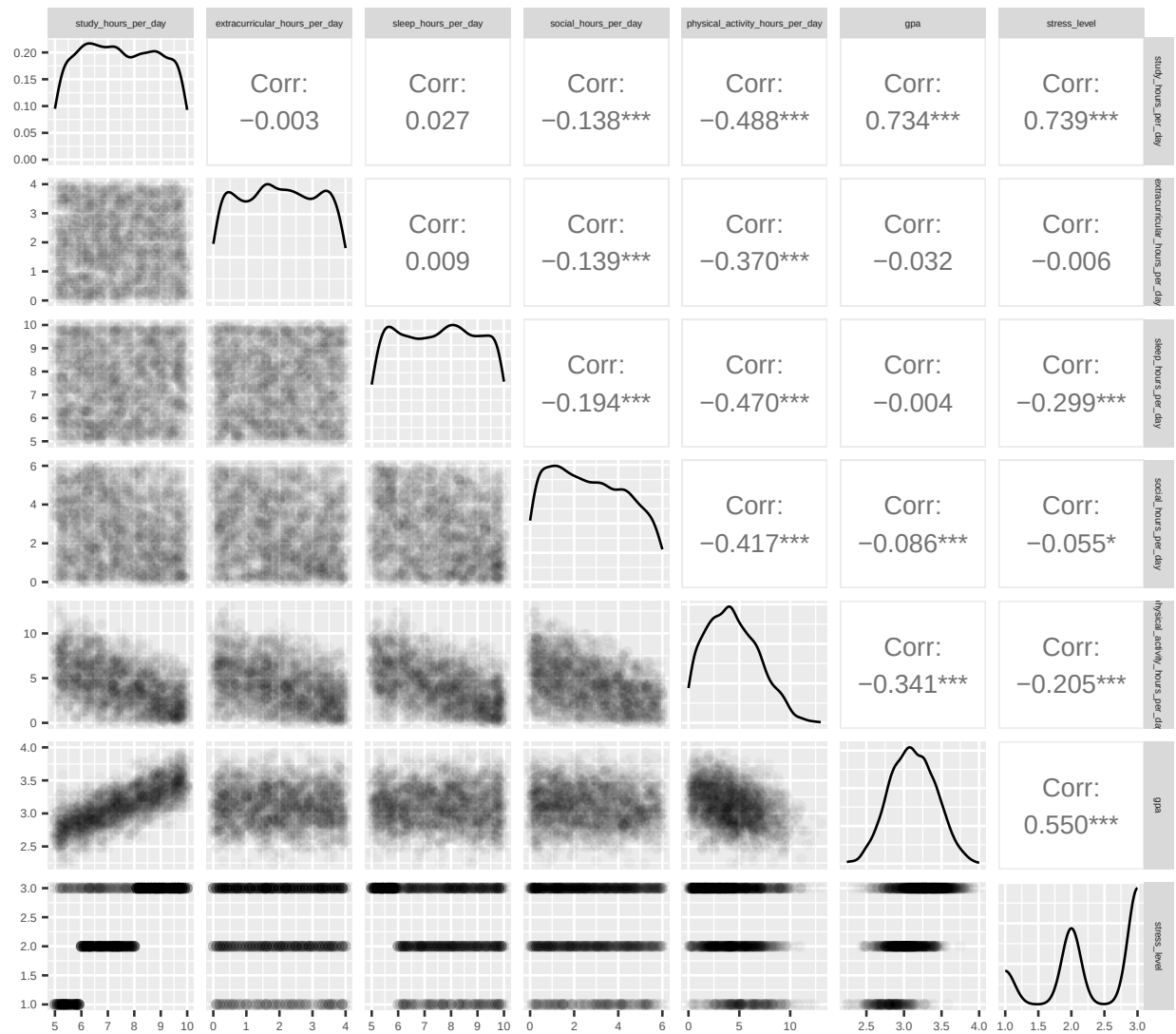


Figure 1: Pairs plot for numeric variables.

```
dat |>
  dplyr::select(where(is.numeric)) |>
  tidyr::pivot_longer(cols = -c(gpa), names_to = "var", values_to = "val") |>
  ggplot(aes(val, gpa)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "loess", se = TRUE) +
  facet_wrap(~ var, scales = "free_x")
```

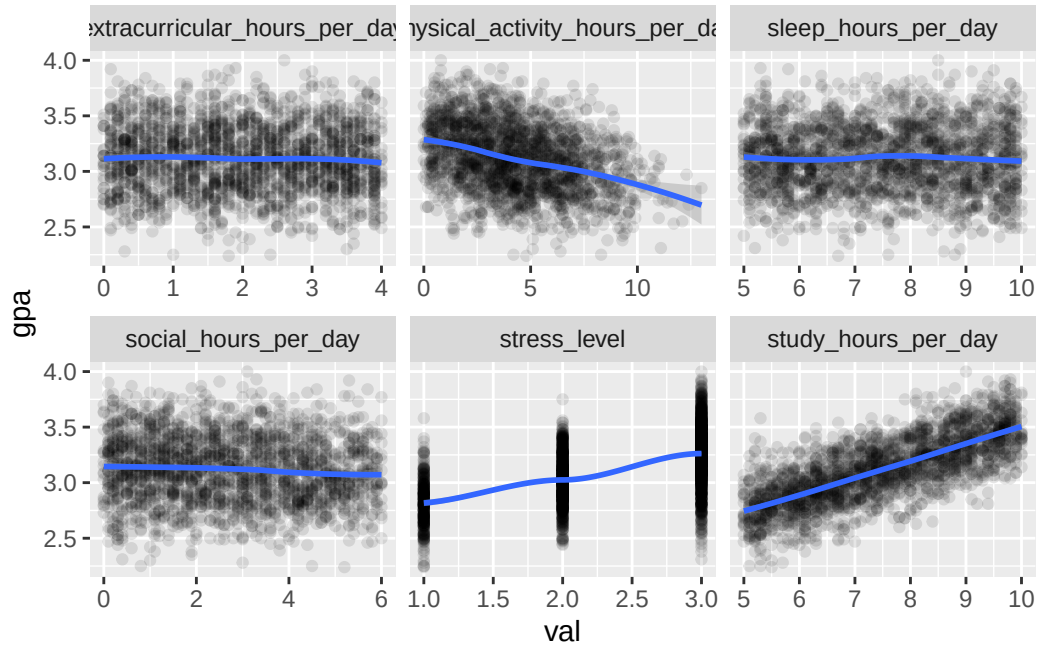


Figure 2: Outcome vs. selected predictors with loess smooth.

1.5 Initial Simple Regressions

```
cand_x <- c("study_hours_per_day", "extracurricular_hours_per_day", "sleep_hours_per_day", "social_hours_per_day", "physical_activity_hours_per_day")
fits <- lapply(cand_x, \(v) lm(reformulate(v, response = "gpa"), data = dat))
models <- tibble::tibble(
  predictor = cand_x,
  glance = purrr::map(fits, broom::glance),
  tidy = purrr::map(fits, broom::tidy)
)

dplyr::bind_rows(purrr::map2(models$predictor, models$glance, \(v, g) dplyr::mutate(g, predictor = v)) |>
  dplyr::select(predictor, r.squared, adj.r.squared, AIC, BIC, p.value) |>
  knitr::kable(digits = 3, caption = "Simple regression comparisons"))
```

Table 3: Simple regression comparisons

predictor	r.squared	adj.r.squared	AIC	BIC	p.value
study_hours_per_day	0.539	0.539	-703.501	-686.698	0.000
extracurricular_hours_per_day	0.001	0.001	845.066	861.869	0.150
sleep_hours_per_day	0.000	0.000	847.101	863.903	0.848
social_hours_per_day	0.007	0.007	832.402	849.205	0.000
physical_activity_hours_per_day	0.116	0.116	599.670	616.472	0.000

1.6 Early Findings

After analyzing the initial summary statistics, pair plots, and the initial simple regressions, we have found the following:

- Study hours per day: Strongest predictor of GPA ($R^2 \approx 0.54, p < 0.001$). There is clear evidence that the number of study hours is related to higher GPA.
- Physical activity hours per day: Moderate predictor of GPA ($R^2 \approx 0.12, p < 0.001$). This explanatory variable suggests a moderate positive effect.
- Social hours per day: Weak predictor of GPA ($R^2 \approx 0.007$), but statistically significant ($p < 0.001$). This explanatory variable has a small effect in practice.
- Extracurricular hours per day: Very weak predictor of GPA ($R^2 \approx 0.001, p \approx 0.15$). This variable likely does not have an effect on GPA.
- Sleep hours per day: Non-significant predictor of GPA ($R^2 \approx 0.000, p \approx 0.85$). This variable likely does not have an effect on GPA.

1.7 Next Steps

Our next steps for this data analysis would be to run models with multiple predictors to test interactions and see how they may be correlated. Additionally, check for nonlinear effects for low predictor variables, for example, testing if sleep helps up to a point but then levels off. Lastly, re-examine sleep specifically to see if the current “no effect” result is accurate or hidden by other factors, outliers, etc.

Overall, the number of study hours was the strongest predictor of GPA, with physical activity showing some benefit as well. Sleep did not appear significant in our simple model, but other research, points to sleep being important. This result was surprising and we plan to explore it further to confirm or challenge our current result. Our next steps will focus on building more complete models that test combinations of predictors, nonlinear patterns, and interactions to get a clearer picture of what drives GPA in college students.

1.8 Reproducibility Snapshot

```
sessionInfo()
```

```
R version 4.5.0 (2025-04-11)
```

```
Platform: aarch64-apple-darwin20
```

```
Running under: macOS Sonoma 14.3
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LA
```

locale:

[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Denver

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] kableExtra_1.4.0 knitr_1.50 patchwork_1.3.0 broom_1.0.8
[5] GGally_2.2.1 skimr_2.2.1 janitor_2.2.1 lubridate_1.9.4
[9] forcats_1.0.0 stringr_1.5.1 dplyr_1.1.4 purrr_1.0.4
[13] readr_2.1.5 tidyr_1.3.1 tibble_3.2.1 ggplot2_3.5.2
[17] tidyverse_2.0.0

loaded via a namespace (and not attached):

[1] gtable_0.3.6 xfun_0.52 lattice_0.22-6 tzdb_0.5.0
[5] vctr_0.6.5 tools_4.5.0 generics_0.1.4 parallel_4.5.0
[9] pacman_0.5.1 pkgconfig_2.0.3 Matrix_1.7-3 RColorBrewer_1.1-3
[13] lifecycle_1.0.4 compiler_4.5.0 farver_2.1.2 textshaping_1.0.1
[17] repr_1.1.7 snakecase_0.11.1 htmltools_0.5.8.1 yaml_2.3.10
[21] pillar_1.10.2 crayon_1.5.3 nlme_3.1-168 ggstats_0.9.0
[25] tidyselect_1.2.1 digest_0.6.37 stringi_1.8.7 labeling_0.4.3
[29] splines_4.5.0 fastmap_1.2.0 grid_4.5.0 cli_3.6.5
[33] magrittr_2.0.3 base64enc_0.1-3 withr_3.0.2 scales_1.4.0
[37] backports_1.5.0 bit64_4.6.0-1 timechange_0.3.0 rmarkdown_2.29
[41] bit_4.6.0 hms_1.1.3 evaluate_1.0.3 viridisLite_0.4.2
[45] mgcv_1.9-1 rlang_1.1.6 Rcpp_1.0.14 glue_1.8.0
[49] xml2_1.3.8 svglite_2.2.1 rstudioapi_0.17.1 vroom_1.6.5
[53] jsonlite_2.0.0 R6_2.6.1 plyr_1.8.9 systemfonts_1.2.3